FinQA Numeric-Answer Parsing Rule — Validation for Sign-off

1. Rule specification

`src/scoring/finqa_numeric_scorer.py` (`FinqaNumericScorer`)

1. **Extraction** — take the `**Final Answer:**` section if present, else the full answer. Select the last numeric token, preferring one carrying a unit word (`%`, `million`, `billion`, `thousand`). A trailing bare `1` after `to/:` is ignored (ratio "38.6 to 1" → 38.6).
2. **Normalization** — strip `$` and thousands separators. `%` → `/100`. `million/billion/thousand` → ×1e6/1e9/1e3 as base.
3. **Scale ambiguity (table units)** — gold `exe_ans` is in raw table units, which vary per table (raw / thousands / millions). Candidates = `{base, base×1e-3, base×1e-6}`. Plain numbers with no unit word are NOT scaled (avoids false positives such as 1,056,000 → 1056).
4. **Comparison** — match if any candidate is within `rtol=0.005` (`math.isclose`, `atol=1e-6`) of `exe_ans`. This covers rounding (NKE 81.37 → 0.81372) and %-vs-decimal (81.37% → 0.8137).
5. **Yes/No questions** — when `exe_ans` is `"yes"`/`"no"`, exact case-insensitive token match on the answer (handles P4 `VERDICT: approved Yes`).

2. Validation — 10 known cases

case	gold `exe_ans`	gold `answer`	model parsed	match	category
NKE/2009	0.81372	81%	81.37%	1.0	% + rounding
ABC/2005	0.9	90%	90.0%	1.0	%
MRO/2013	0.24744	24.7%	24.7%	1.0	% + trailing year
AAL/2016	0.13202	13.2%	13.2%	1.0	% rounding
TMUS/2017	82352.94118	82352.94	82,352.94	1.0	plain number
MRO/2003	1056.0	1056	1,056	1.0	thousand unit
AMT/2014	33458.0	63458*	33.458M → 33458	1.0	million in table-thousands
ETFC/2011	4.6	4.6	38.6 (to 1)	0.0	ratio tail (model wrong)
AMT/2003	0.33333	33.3%	10,000 sqft	0.0	model omits the %
JKHY/2019	"no"	no	no	1.0	yes/no

Case → full `query_id` mapping: NKE/2009 → `NKE/2009/page_43.pdf-2`, ABC/2005 → `ABC/2005/page_49.pdf-4`, MRO/2013 → `MRO/2013/page_19.pdf-4`, AAL/2016 → `AAL/2016/page_8.pdf-1`, TMUS/2017 → `TMUS/2017/page_29.pdf-2`, MRO/2003 → `MRO/2003/page_45.pdf-2`, AMT/2014 → `AMT/2014/page_149.pdf-2`, ETFC/2011 → `ETFC/2011/page_144.pdf-2`, AMT/2003 → `AMT/2003/page_27.pdf-1`, JKHY/2019 → `JKHY/2019/page_18.pdf-3`.

* FinQA quirk: human-typed `answer` and program-executed `exe_ans` differ here. Per Álvaro's ruling the gold is `exe_ans` (33458).

8/10 match; the 2 mismatches are genuine model errors, not parser errors.

3. Full-run summary (573 records, current telemetry)

pattern	numeric accuracy
1	80/150 = 0.533
2	73/150 = 0.487
3	73/150 = 0.487
4	79/123 = 0.642
total	**305/573 = 0.532**

Note: pattern 4's higher score correlates with the `VERDICT:` wrapper emitting a clean final number — flagged for the report discussion, not a parser artifact.

4. Open items for sign-off

1. Confirm `rtol=0.005` tolerance.
2. Confirm scale-candidate set `{base, base×1e-3, base×1e-6}` (only when a unit word is present).
3. Confirm yes/no handled as exact token match.
4. Confirm using `exe_ans` (not `answer`) as gold even when they disagree (AMT/2014 33458 vs 63458).

5. Files

- `src/scoring/finqa_numeric_scorer.py` — scorer
- `tests/test_finqa_numeric_scorer.py` — 13 unit tests (all pass)
- `telemetry/a100/finqa_references.jsonl` — 1147 `query_id → exe_ans` from pinned `czyssrs/FinQA` `test.json` (commit `0f16e2867befa6840783e58be38c9efb9229d742`)